OKBA BOUZIANE

Full-Stack & AI Engineer

Algiers, Algeria • +213 784 558 472 • okba.bouziane@outlook.com
devokba.vercel.app • linkedin.com/in/okba-bouziane • github.com/okba-bouziane

TECHNICAL SKILLS

AI & LLM: LangGraph, LangChain, LlamaIndex, AI Agents, RAG, Graph RAG, MCP, PGVector, Prompt Engineering, LLM APIs
Frontend: Angular, React, TypeScript, RxJS, React Native
Backend: Node.js, NestJS, Express.js, Python, FastAPI, REST, GraphQL, gRPC
Databases: PostgreSQL, MongoDB, Neo4j, Redis, MinIO (S3)
Security & DevOps: Keycloak, OAuth2 / OpenID Connect, RBAC, Docker, CI/CD, Git
AI Tools: Claude Code, Cursor, Windsurf, Kiro

PERSONAL PROJECTS

Secure Agent Workspace | React, NestJS, LangGraph, Keycloak, PostgreSQL
An AI assistant that works on company data with the permissions of the user who is logged in.

- The agent can only do what the user is allowed to do, and every change waits for a human to approve it.
- Answers questions from company documents with sources, blocks prompt injection, and has 82 automated tests.

VoiceDesk | Python, FastAPI, LiveKit, LangGraph, gRPC, PostgreSQL
A voice receptionist you talk to in the browser, live online.

- Books, lists and cancels appointments by voice, and answers questions from a knowledge base.
- Hands the call to a human when needed; bookings run in a separate gRPC microservice.

LexGraph | Python, FastAPI, LlamaIndex, Neo4j, LangGraph
A contract search assistant that follows the links between documents.

- Stores contracts, amendments and guarantees as a Neo4j knowledge graph to answer multi-document questions.
- On those questions it finds 87% of the needed documents versus 72% for plain vector search; covered by 41 automated tests.

PROFESSIONAL EXPERIENCE

Full-Stack Developer | Smartest Algeria, Algiers — Nov 2022 – Oct 2026

- Corrosion risk on an oil & gas well-integrity platform was reviewed one well at a time; built a LangGraph AI agent that checks each well against the inspection and corrosion history of nearby wells, flagging at-risk wells that single-well reviews missed.
- The platform's UI had recurring bugs; rebuilt its Angular front end on the shared component library to fix them, and built Node.js microservices for well-data ingestion and integrity assessment as one of 4 backend developers.
- Teams processed documents by hand; added LLM summarization, smart search and form auto-fill, cutting manual data entry.
- Each app had its own login; set up Keycloak SSO (OAuth2/OIDC, JWT, RBAC), giving users one secure login for all apps.
- Each app handled licensing on its own; built a central SaaS licensing service that every Smartest app now uses.
- Users needed a developer for every new report; built a Word-like report designer so business users create their own reports.
- Several apps needed flow editing; built flow-dcl, a reusable Angular flow-editor library, so no app had to rebuild it.
- Legacy modules were tightly coupled; refactored them to Clean Architecture with unit tests, making releases safer.
- Reporting screens were slow; optimized the PostgreSQL queries behind them, speeding up the heaviest screens.

Mobile & Web Developer | Tebessa Innoest, Tebessa — Mar 2020 – Jan 2022

- Incubated founders had ideas but no product to pitch; turned them into mobile and web MVPs in short delivery cycles, advising on technical scope and feature priority so each team could demo a working first version.
- Start-ups needed apps on both platforms on a small budget; built cross-platform React Native apps (iOS and Android) with authentication, REST APIs, push notifications and offline-friendly screens, plus React websites for demo days.

EDUCATION

MSc, Artificial Intelligence | Badji Mokhtar University, Annaba — 2020 – 2022
BSc, Computer Science | Badji Mokhtar University, Annaba — 2017 – 2020

ADDITIONAL INFORMATION

Languages: Arabic (Native), English (Full Professional), French (Limited Working) | **Military Service:** Exempt